CLASS 9 · PART TEST-1 · STUDENT PERFORMANCE REPORT

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Pragya G9 (EXT) Achievers · Adm No **246419816**

Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

166 / 296 — **56.1%**

122	87	9/79	16
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **183.7 / 300** (61.2%), 5 of 6 tests → this test **56.1%** ▼ **5.1**

130 marks lost: 110 on 22 wrong answers (incl. 22 negative) · 20 on 5 skipped

JEE Advanced

09 Oct 2026 · 50 questions

69 / 176 — **39.2%**

235	168	20/78	32†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **104.0 / 180** (57.8%), 5 of 6 tests → this test **39.2%** ▼ **18.6**

107 marks lost: 91 on 25 wrong answers (incl. 6 negative) · 8 on 2 skipped · 8 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 79 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 166 / 296 · 56.1%							
Maths	62/100	62%	180	43.1 / 95	75.8/100 75.8%	▼ 13.8	17 / 6 / 2
Physics	37/100	37%	265	21.9 / 90	66.4/100 66.4%	▼ 29.4	12 / 11 / 2
Chemistry	67/96	70%	46	41.4 / 81	41.1/100 41.1%	▲ 28.7	18 / 5 / 1
JEE Advanced 69 / 176 · 39.2%							
Maths	36/60	60%	182	24.5 / 60	43.3/60 72.2%	▼ 12.2	11 / 5 / 1
Physics	16/60	27%	325	12.3 / 56	35.6/60 59.3%	▼ 32.6	5 / 11 / 1
Chemistry	17/56	30%	561	15.8 / 37	24.6/60 41.1%	▼ 10.7	7 / 9 / 0

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Chemistry** is your strongest JEE Mains subject: 67/96 (70%), All-India subject rank 46.
2. **Physics** needs the most work in JEE Mains: 37/100 (37%), subject rank 265; you lost 13 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 36/60 (60%), All-India subject rank 182.
4. **Physics** needs the most work in JEE Advanced: 16/60 (27%), subject rank 325; you lost 12 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Advanced Physics** – 27% here against 59% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 70% here against 41% on average.
7. **1 question** you lost were answered correctly by 70%+ of the batch's top 100 – about **6 marks** within easy reach (Advanced Q39).
8. Negative marking cost you **22 in JEE Mains and 6 in JEE Advanced**.
9. You attempted almost everything (only 7 questions skipped across both papers), so the marks to gain are in **accuracy**: wrong answers – 22 in JEE Mains and 25 in JEE Advanced.

JEE Mains – question by question

166 / 296 · 56.1%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Physics	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
Chemistry	51	52	53	54	55	56	57	~~58~~	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 79 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40-69% · **Tough one** below 40%

Maths

8 of 25 questions lost · **38** marks · 5 within reach · 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q13	Within reach	**Trigonometric Equations** – Solutions of tan x + sec x = 2 cos x in [0, 2π]	Single correct · +4 / −1	Skipped	4	66%	58%
Q12	Within reach	**Inverse Trigonometric Functions**	Single correct · +4 / −1	Wrong	5	64%	52%
Q15	Within reach	**Functions** – Inverse of f(x) = x² + x + 1 on a restricted domain	Single correct · +4 / −1	Skipped	4	63%	44%
Q3	Within reach	**Functions**	Single correct · +4 / −1	Wrong	5	58%	53%
Q1	Within reach	**Functions**	Single correct · +4 / −1	Wrong	5	54%	41%
Q7	Tough one	**Trigonometric Equations** – √3 sin πx + cos πx = quadratic – number of solutions	Single correct · +4 / −1	Wrong	5	37%	16%
Q17	Tough one	**Inverse Trigonometric Functions**	Single correct · +4 / −1	Wrong	5	34%	35%
Q20	Tough one	**Inverse Trigonometric Functions** – Composite inverse-trig function – counting solutions	Single correct · +4 / −1	Wrong	5	19%	13%

Physics

13 of 25 questions lost · **63** marks · ● 2 within reach · ● 11 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q40	Within reach	**Laws of Motion**	Single correct · +4 / −1	Wrong	5	63%	46%
Q37	Within reach	**Laws of Motion** – Tension in a pulley–block system	Single correct · +4 / −1	Wrong	5	49%	43%
Q36	Tough one	**Laws of Motion**	Single correct · +4 / −1	Wrong	5	39%	24%
Q27	Tough one	**Circular Motion** – Coins on a rotating disc – friction limit at each radius	Single correct · +4 / −1	Wrong	5	37%	23%
Q47	Tough one	**Laws of Motion**	Numerical · +4 / −1	Skipped	4	36%	24%
Q35	Tough one	**Circular Motion** – Power delivered by the centripetal force	Single correct · +4 / −1	Wrong	5	34%	18%
Q31	Tough one	**Work, Energy & Power** – KE vs height for a stone thrown up from a cliff	Single correct · +4 / −1	Wrong	5	32%	10%
Q42	Tough one	**Work, Energy & Power** – Change in PE of a ball thrown from a tower	Single correct · +4 / −1	Wrong	5	32%	15%
Q26	Tough one	**Momentum & Collisions** – KE loss in head-on collision from relative velocities	Single correct · +4 / −1	Skipped	4	31%	22%
Q38	Tough one	**Work, Energy & Power** – Work along different paths; conservative or not	Single correct · +4 / −1	Wrong	5	25%	13%
Q43	Tough one	**Laws of Motion** – Tension in a spring pulled by 5 N at each end	Single correct · +4 / −1	Wrong	5	24%	10%
Q28	Tough one	**Friction** – Friction vs incline angle graph (angle of repose)	Single correct · +4 / −1	Wrong	5	21%	11%
Q45	Tough one	**Work, Energy & Power** – Speed on hitting the ground – energy conservation	Single correct · +4 / −1	Wrong	5	18%	6%

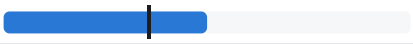
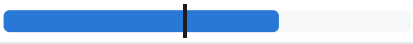
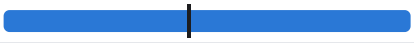
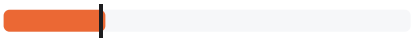
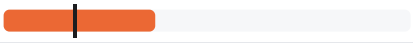
Chemistry

6 of 24 questions lost · **29** marks · ● 1 within reach · ● 5 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q59	Within reach	**Isomerism**	Single correct · +4 / −1	Skipped	4	43%	38%
Q73	Tough one	**General Organic Chemistry**	Numerical · +4 / −1	Wrong	5	32%	23%
Q74	Tough one	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	Wrong	5	20%	11%
Q63	Tough one	**Hydrocarbons** – Monochlorination products incl. stereoisomers	Single correct · +4 / −1	Wrong	5	16%	20%
Q75	Tough one	**General Organic Chemistry** – Bases stronger than aniline	Numerical · +4 / −1	Wrong	5	11%	9%
Q71	Tough one	**Isomerism** – Count of optically active compounds	Numerical · +4 / −1	Wrong	5	6%	10%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Functions	7	14/28		35%	Q1, Q3, Q15
Inverse Trigonometric Functions	10	25/40		48%	Q12, Q17, Q20
Trigonometric Equations	7	19/28		44%	Q7, Q13
Sequences & Series	1	4/4		45%	– none
Physics					
Work, Energy & Power	5	0/20		9%	Q31, Q38, Q42, Q45
Laws of Motion	8	8/32		23%	Q36, Q37, Q40, Q43, Q47
Friction	2	3/8		17%	Q28
Circular Motion	7	18/28		30%	Q27, Q35
Momentum & Collisions	3	8/12		25%	Q26
Chemistry					
General Organic Chemistry	10	25/40		32%	Q73, Q74, Q75
Hydrocarbons	4	11/16		42%	Q63
Isomerism	10	31/40		55%	Q59, Q71

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

8 questions · easy misses first
cropped from the question paper

Q13 — Trigonometric Equations – Solutions of tan x + sec x = 2 cos x in [0, 2π]

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **66%** got it right · Campus: **58%** (46 of 79)

Within reach
You: Skipped

13. Number of solution(s) of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$ is

A) 1 B) 2 C) 3 D) 4

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12 — Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **64%** got it right · Campus: **52%** (41 of 79)

Within reach
You: Wrong

12. Let 'S' be the complete set of solutions of the inequation $\left(\sec^{-1} x\right)^2 - 5\left(\sec^{-1} x\right) + 6 > 0$

Which of the following intervals is not contained in the solution set?

A) $(\sec 2, \sec 3)$ B) $(-\infty, \sec 2)$ C) $(\sec 3, -1]$ D) $[1, \infty)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15 — Functions – Inverse of f(x) = x² + x + 1 on a restricted domain

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **63%** got it right · Campus: **44%** (35 of 79)

Within reach
You: Skipped

15. Let $f:\left(-\infty, \frac{-1}{2}\right] \to \left[\frac{3}{4}, \infty\right)$, $f(x) = x^2 + x + 1$. Then $f^{-1}(x) = \ldots\ldots\ldots$

A) $\frac{-1 \pm \sqrt{4y-3}}{2}$ B) $\frac{-1 \pm \sqrt{4x-3}}{2}$

C) $\frac{-1 + \sqrt{4x-3}}{2}$ D) $\frac{-1 - \sqrt{4x-3}}{2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q3

Functions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **58%** got it right · Campus: **53%** (42 of 79)

Within reach
You: Wrong

3. Consider the statements:

S_1 : For any two functions $f(x)$ and $g(x)$, let range of $f(x)$ be $[a,b]$ and range of $g(x)$ be $[c,d]$, then range of $f(x)+g(x)$ is $[a+c,b+d]$

S_2 : Range of $h(x)=\sin^2 x+\cos^2 x$ is $[0,2]$

A) Only S_1 is true
B) Only S_2 is true
C) Both Statements are false
D) Both statements are true

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q1

Functions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **54%** got it right · Campus: **41%** (32 of 79)

Within reach
You: Wrong

1. The function $f: R-\{0\}\to R, f(x)=\dfrac{x}{e^x-1}+\dfrac{x}{2}+1$ is

A) even
B) odd
C) One- one
D) periodic

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q7

Trigonometric Equations – √3 sin πx + cos πx = quadratic – number of solutions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **37%** got it right · Campus: **16%** (13 of 79)

Tough one
You: Wrong

7. Number of solutions of $\sqrt{3}\sin(\pi x)+\cos(\pi x)=x^2-\dfrac{2}{3}x+\dfrac{19}{9}$ is .....

A) 0
B) 1
C) 2
D) 3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17 Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **34%** got it right · Campus: **35%** (28 of 79)

Tough one

You: Wrong

17. Let $\left(\sin^{-1}x\right)^2+\left(\cos^{-1}x\right)^2=k\left(where\ \ x\in[-1,1]\right)$ then k cannot be

A) $\frac{\pi^2}{8}$ B) $\frac{\pi^2}{3}$ C) $\frac{\pi^2}{9}$ D) π^2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20 Inverse Trigonometric Functions – Composite inverse-trig function – counting solutions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **19%** got it right · Campus: **13%** (10 of 79)

Tough one

You: Wrong

20. Consider the functions $rage(x)=\sin^{-1}\sqrt{x}+e^{\left(\tan^{-1}\left(\sqrt{x-1}\right)\right)^2}+\left(\cos^{-1}\left(\sqrt{x}\right)\right)^3$ and $bait(x)=x^2+x+1$ within their respective domains. Then the number of solutions of $rage\left(bait(x)\right)<3$ is/are .....

A) 0 B)1 C)2 D)3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

13 questions · easy misses first
cropped from the question paper

Q40 — Laws of Motion

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **63%** got it right · Campus: **46%** (36 of 79)

Within reach
You: Wrong

40. Statement – I: Force required to move a body uniformly along straight line is can be zero.

Statement – II: The force required to move a body uniformly along the circle is zero.

A) Statement – I is true , Statement – II false

B) Both Statements are true

C) Both Statements are false

D) Statement – I false , Statement – II true

Why did I lose this question? (write the reason and the correct approach)

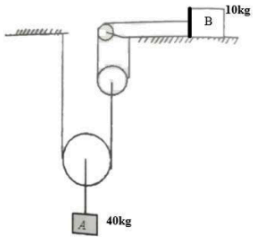
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q37 — Laws of Motion – Tension in a pulley-block system

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **49%** got it right · Campus: **43%** (34 of 79)

Within reach
You: Wrong

37. Fig shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then $\left(g = 10 m/s^2\right)$(Mass of a is 40Kg and B is 10Kg on smooth surface). The tension in the string connecting the block B is



A) $20N$ B) $80N$ C) $60N$ D) $40N$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36 Laws of Motion

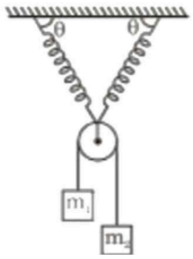
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **39%** got it right · Campus: **24%** (19 of 79)

Tough one

You: Wrong

36. Two identical ideal springs of spring constant (K) 1000N/m as connected by an ideal pulley as shown and system is arranged in vertical plane. Δt equilibrium θ is 60° and masses m_1 and m_2 are 2kg and 3kg respectively. Then elongation in each spring when $\theta = 60^o$ is $\left(g = 10m/s^2\right)$



A) $x = 1.6cm$ B) $x = 4.8cm$ C) $x = 4.8\sqrt{3}$ D) $x = 1.6\sqrt{3}cm$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27 Circular Motion – Coins on a rotating disc – friction limit at each radius

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **37%** got it right · Campus: **23%** (18 of 79)

Tough one

You: Wrong

27. A disc revolves with a speed of 100/3 rev/min and has radius of 15cm. Two coins of same mass as disc are placed at 4 cm and 14cm away from the centre of disc. If the coefficient of friction between the coins and the disc is 0.15, which of the two coins will revolve with disc?
A) both the coins will revolve
B) No coin will revolve with disc
C) coin at 4cm will revolve with disc
D) coin at 14cm will revolve with disc

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Laws of Motion

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **36%** got it right · Campus: **24%** (19 of 79)

Tough one

You: Skipped

47. A body with mass 5kg is acted upon by a force $\vec{F}=\left(-3\hat{i}+4\hat{j}\right)N$. If its initial velocity at $t=0\text{sec}$ is $u=\left(6\hat{i}-12\hat{j}\right)m/s$. The time at which it will just have a velocity along y-axis .....sec

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35

Circular Motion – Power delivered by the centripetal force

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **34%** got it right · Campus: **18%** (14 of 79)

Tough one

You: Wrong

35. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration changing with time 't' as $a_c=k^2t^2$, where K is a constant. The power delivered to the particle by centripetal force acting on it is

A) zero B) mk^2rt C) mk^2r^2t D) $mk^2\sqrt{rt}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Work, Energy & Power – KE vs height for a stone thrown up from a cliff

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

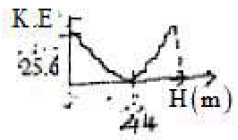
Top 100: **32%** got it right · Campus: **10%** (8 of 79)

Tough one

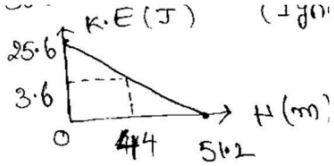
You: Wrong

31. A stone of mass 50g is thrown vertically upwards with a velocity of 12m/s from edge of a cliff 44m high. Identify kinetic energy of stone versus height from bottom of cliff $\left(g = 10m/s^2\right)$(ignore air resistance)

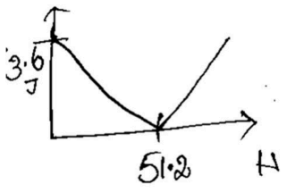
A)



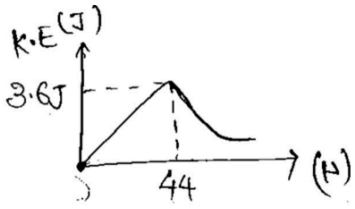
B)



C)



D)



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q42

Work, Energy & Power – Change in PE of a ball thrown from a tower

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **32%** got it right · Campus: **15%** (12 of 79)

Tough one

You: Wrong

42. At t =0 sec a 1kg ball is thrown from top to a tall tower with velocity $\vec{V} = \left(8\hat{i} + 9\hat{j}\right)$ m/s. The change in potential energy of ball – earth system between $t = 0$ *to* $t = 5$sec is

$\left(g = 10m/s^2\right)$

A) 800J B) 1536J C) 2100J D) -800J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26 **Momentum & Collisions** – KE loss in head-on collision from relative velocities

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **31%** got it right · Campus: **22%** (17 of 79)

Tough one

You: Skipped

26. Two bodies of masses 3 kg and 2 kg collide head on. Their relative velocities before and after collision are 15m/s and 5 m/s respectively. The loss of kinetic energy of system is

A) 120J B) 100J C) 80J D) 240J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q38 **Work, Energy & Power** – Work along different paths; conservative or not

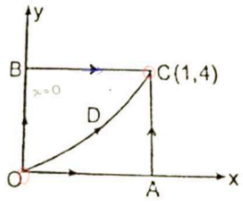
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **25%** got it right · Campus: **13%** (10 of 79)

Tough one

You: Wrong

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2 y\hat{j}$ on the particle along



Column – I	**Column – II**
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43

Laws of Motion – **Tension in a spring pulled by 5 N at each end**

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **24%** got it right · Campus: **10%** (8 of 79)

Tough one

You: Wrong

43. The tension in the massless spring is



A) zero B) 10N C) 2.5N D) 5N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Friction – **Friction vs incline angle graph (angle of repose)**

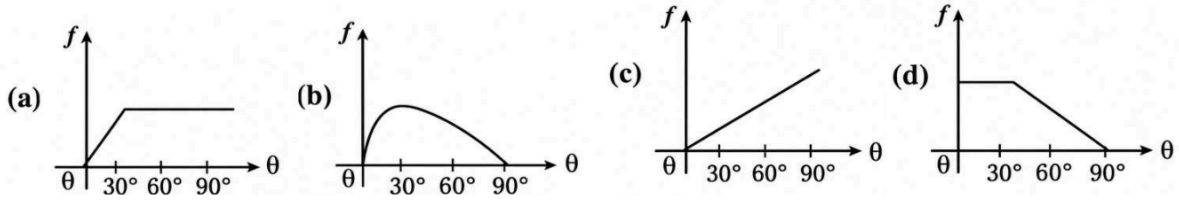
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **21%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Wrong

28. A block is placed on a rough plane whose inclination to the horizontal (θ) can be varied. The angle of repose is 30°. The graph that correctly indicate the variation of frictional force (f) between block and the plane with θ is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45 **Work, Energy & Power** – Speed on hitting the ground – energy conservation

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **18%** got it right · Campus: **6%** (5 of 79)

Tough one

You: Wrong

45. A cricket bowler releases the ball in two different ways

i) giving it only horizontal velocity and

ii) giving it horizontal velocity and small downward velocity

The speeds at the time release is same. Both are released at a height H from ground then which one will have greater speed when ball hits the ground?(Neglect air resistance)

A) Ist way B) IInd way C) same in two ways D) None of the above

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

6 questions · easy misses first
cropped from the question paper

Q59 **Isomerism**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **43%** got it right · Campus: **38%** (30 of 79)

Within reach
You: Skipped

59. Which of the following statements regarding 1,2-dimethylcyclopropane (I) and 1,2-dimethyl cyclobutane (II) are wrong ?

A) Both of them show three stereoisomers each

B) The cis form of both is optically inactive (meso) and the trans form of both has a pair of enantiomers

C) The cis form of both has a pair of enantiomers and the trans form of both is optically inactive (meso)

D) The meso form of both is optically inactive due to the presence of plane of symmetry.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q73

General Organic Chemistry

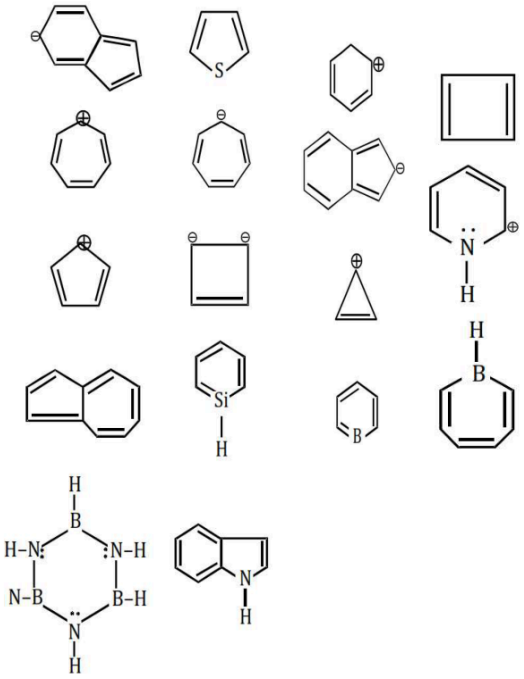
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **32%** got it right · Campus: **23%** (18 of 79)

Tough one

You: Wrong

73. How Many of the following are Aromatic compounds?



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74

General Organic Chemistry – Compounds giving a positive bicarbonate test

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Wrong

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q63

Hydrocarbons – Monochlorination products incl. stereoisomers

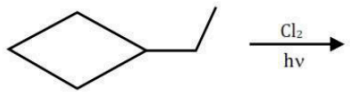
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **16%** got it right · Campus: **20%** (16 of 79)

Tough one

You: Wrong

63. Number of monochlorinated product when following compound undergo reaction with $Cl_2 / h\nu$ (including stereo isomers)



A) 5 B) 9 C) 10 D) 6

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q75

General Organic Chemistry – Bases stronger than aniline

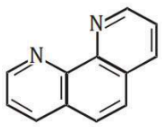
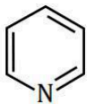
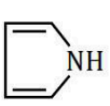
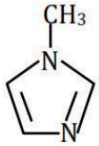
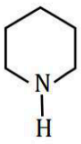
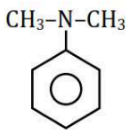
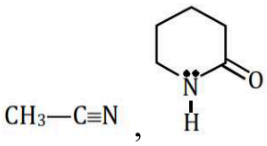
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **11%** got it right · Campus: **9%** (7 of 79)

Tough one

You: Wrong

75. How many of the following are more basic then Aniline?

NH_3, $PhCH_2NH_2$,  ,  ,  ,  ,  ,  , $CH_3—CH_2—NH_2$, 

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q71 **Isomerism – Count of optically active compounds**

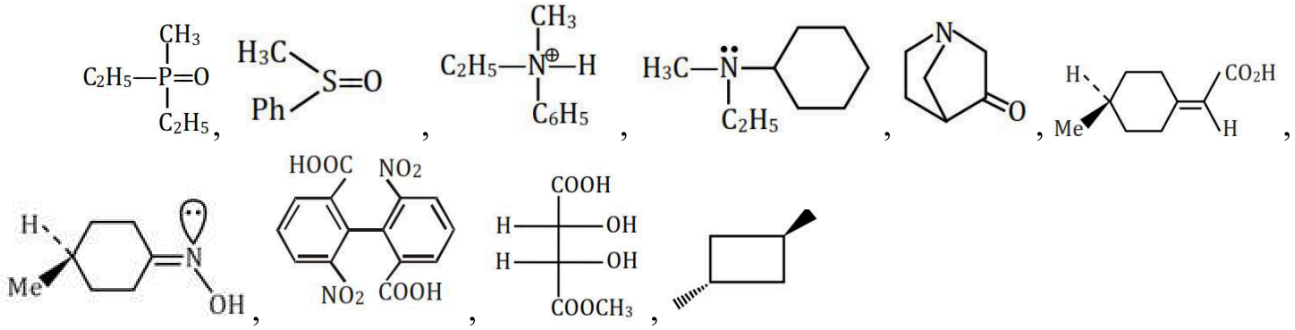
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **6%** got it right · Campus: **10%** (8 of 79)

Tough one

You: Wrong

71. Total no of compounds which are optically active are _____



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced – question by question

69 / 176 · 39.2%
09 Oct 2026

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Physics	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
Chemistry	35	36	37	38	39	40	41	42	43	44	~~45~~	46	47	48	49	50	51

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 78 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40-69% · **Tough one** below 40%

Maths

6 of 17 questions lost · 21 marks (+3 on partly-correct answers) · 4 within reach · 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q11	Within reach	Functions	Integer · +4 / 0	Skipped	4	62%	45%
Q15	Within reach	Inverse Trigonometric Functions	Paragraph · +3 / 0	Wrong	3	57%	53%
Q9	Within reach	Inverse Trigonometric Functions	Integer · +4 / 0	Wrong	4	54%	58%
Q16	Within reach	Trigonometric Equations	Paragraph · +3 / 0	Wrong	3	51%	42%
Q8	Tough one	**Functions** – Period from two lines of symmetry	Integer · +4 / 0	Wrong	4	12%	13%
Q17	Tough one	**Trigonometric Equations** – Sum of solutions of $\cos(\pi\sqrt{(x-100)})\cdot\cos(\pi\sqrt{x}) = 1$	Paragraph · +3 / 0	Wrong	3	6%	6%

Physics

12 of 17 questions lost · 44 marks · 3 within reach · 9 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q25	Within reach	Laws of Motion	Integer · +4 / 0	Wrong	4	47%	44%
Q31	Within reach	**Friction** – Friction on a block at rest inside an accelerating cube	Paragraph · +3 / 0	Wrong	3	45%	51%
Q19	Within reach	Circular Motion	Single correct · +3 / −1	Wrong	4	40%	38%
Q20	Tough one	**Work, Energy & Power** – Maximum spring extension – energy method	Single correct · +3 / −1	Wrong	4	29%	27%
Q26	Tough one	Circular Motion	Integer · +4 / 0	Wrong	4	24%	17%
Q29	Tough one	**Circular Motion** – Net acceleration in a vertical circle at 37°	Integer · +4 / 0	Wrong	4	19%	14%
Q30	Tough one	**Friction** – Work by friction seen from an accelerating observer	Integer · +4 / 0	Wrong	4	19%	12%
Q18	Tough one	**Friction** – Static vs kinetic friction under a time-varying force	Single correct · +3 / −1	Wrong	4	16%	9%
Q34	Tough one	**Work, Energy & Power** – Work by conservative forces from the U-x graph	Paragraph · +3 / 0	Wrong	3	12%	9%
Q27	Tough one	**Friction** – Net work done by friction between sliding blocks	Integer · +4 / 0	Skipped	4	8%	10%
Q32	Tough one	**Friction** – Total force exerted by a block on an accelerating cube	Paragraph · +3 / 0	Wrong	3	8%	6%
Q33	Tough one	**Work, Energy & Power** – Work-energy theorem from a-t and U-x graphs	Paragraph · +3 / 0	Wrong	3	2%	1%

Chemistry

9 of 16 questions lost · 34 marks (+5 on partly-correct answers) · 1 easy miss · 3 within reach · 5 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q39	Easy miss	Hydrocarbons	Multiple correct · +4 / −2	Wrong	6	87%	69%
Q35	Within reach	General Organic Chemistry	Single correct · +3 / −1	Wrong	4	58%	42%
Q44	Within reach	General Organic Chemistry	Integer · +4 / 0	Wrong	4	56%	46%
Q49	Within reach	General Organic Chemistry	Paragraph · +3 / 0	Wrong	3	53%	46%
Q48	Tough one	General Organic Chemistry – Counting electrophiles	Paragraph · +3 / 0	Wrong	3	37%	27%
Q46	Tough one	Isomerism – Count of optically active compounds	Integer · +4 / 0	Wrong	4	15%	15%
Q51	Tough one	Hydrocarbons – Chiral products of free-radical monobromination	Paragraph · +3 / 0	Wrong	3	15%	4%
Q47	Tough one	Hydrocarbons – Dichlorination isomers of C_5H_{10} (VD 35)	Integer · +4 / 0	Wrong	4	7%	5%
Q50	Tough one	Hydrocarbons – Preparation of alkanes – count correct statements	Paragraph · +3 / 0	Wrong	3	4%	3%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Functions	6	≈ 14/22		40%	Q8, Q11
Trigonometric Equations	5	≈ 11/17		35%	Q16, Q17
Inverse Trigonometric Functions	6	≈ 14/21		55%	Q9, Q15
Physics					
Friction	5	≈ -1/17		12%	Q18, Q27, Q30, Q31, Q32
Work, Energy & Power	4	≈ 2/12		22%	Q20, Q33, Q34
Circular Motion	4	≈ 3/15		17%	Q19, Q26, Q29
Laws of Motion	3	≈ 8/12		39%	Q25
Momentum & Collisions	1	≈ 4/4		33%	– none
Chemistry					
Hydrocarbons	5	≈ 2/18		22%	Q39, Q47, Q50, Q51
Isomerism	2	≈ 4/8		16%	Q46
General Organic Chemistry	9	≈ 16/30		43%	Q35, Q44, Q48, Q49

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

6 questions · easy misses first
cropped from the question paper

Q11 Functions

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **62%** got it right · Campus: **45%** (35 of 78)

Within reach
You: Skipped

11. Let $f(x)=|x|$. Let $g(x)=\min(f(t)), t\in(-\infty, x]$
(For eg. $g(-1)=1$ and $g(2)=0$), then $g(-11)+g(19)=?$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15 Inverse Trigonometric Functions

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **57%** got it right · Campus: **53%** (41 of 78)

Within reach
You: Wrong

PARAGRAPH-I

A mischievous student decides to play around with inverse trigonometric functions. She decides that since $f:\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\to[-1,1], f(x)=\sin x$ and $g:[\pi,2\pi]\to[-1,1], g(x)=\cos x$ are invertible , she defines

$h:[-1,1]\to\left[\frac{\pi}{2},\frac{3\pi}{2}\right], h(x)=\sin^{-1}x$

$j:[-1,1]\to[\pi,2\pi], j(x)=\cos^{-1}x$

So that $h=f^{-1}$ and $j=g^{-1}$. (For example under this definition $\sin^{-1}(0)=\pi$ and $\cos^{-1}(0)=\frac{3\pi}{2}$). Under these definitions, answer the following :(use $\pi=3.14$)

15. $\sin^{-1}\left(\sin\frac{\pi}{3}\right)=$.....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q9

Inverse Trigonometric Functions

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **54%** got it right · Campus: **58%** (45 of 78)

Within reach

You: Wrong

9. $\sum_{n=1}^{5} \sin^{-1}\left(\sin\left(2n-1\right)\right) = \ldots\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16

Trigonometric Equations

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **51%** got it right · Campus: **42%** (33 of 78)

Within reach

You: Wrong

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2 - m^2 = k$ then we use $(n+m)(n-m) = k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n,m).

Eg: $n^2 - m^2 = 6 \Rightarrow (n+m)(n-m) = (\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

16. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-4}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is $\ldots\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8

Functions – Period from two lines of symmetry

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **12%** got it right · Campus: **13%** (10 of 78)

Tough one

You: Wrong

8. Let $a \in R$. If $f(a-x) = f(a+x) \forall x \in R$, then $x = a$ is a line of symmetry in the graph of $y = f(x)$. Consider a non-constant function $g(x)$ such that $g(1+x) = g(1-x) \forall x \in R$ and $g(2026+x) = g(2026-x) \forall x \in R$. Then all such $g(x)$ must be periodic. The smallest positive number that could be a period for all such $g(x)$ is $\ldots\ldots\ldots$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17 **Trigonometric Equations** – Sum of solutions of cos(π√(x−100))·cos(π√x) = 1

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Wrong

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2 - m^2 = k$ then we use $(n+m)(n-m) = k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n,m).

Eg: $n^2 - m^2 = 6 \Rightarrow (n+m)(n-m) = (\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

17. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-100}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is .....

Why did I lose this question? (write the reason and the correct approach)

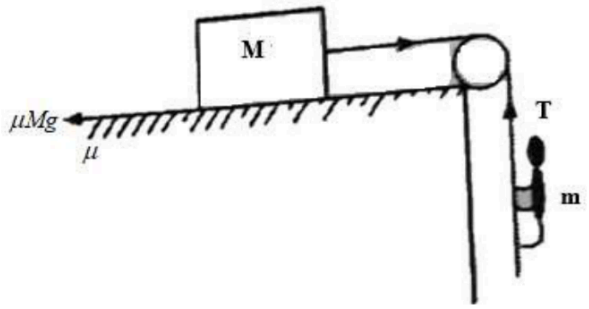
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q25

Laws of Motion
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **47%** got it right · Campus: **44%** (34 of 78)

Within reach
You: Wrong

25. A monkey of mass 'm' climbs up to a rope hung over a fixed pulley with an acceleration relative to rope is $\frac{g}{4} m/s^2$. The opposite end of the rope is tied to a block of mass M lying on a rough horizontal plane. The coefficient of friction between the block and horizontal plane is μ. Tension in rope is $T = \frac{M[5m - 4\mu M]}{k[M+m]} + \mu Mg$ value of k is .....



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Friction – Friction on a block at rest inside an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

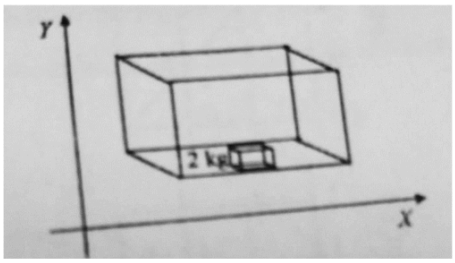
Top 100: **45%** got it right · Campus: **51%** (40 of 78)

Within reach

You: Wrong

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V}=\left(5t\hat{i}+2t\hat{j}\right)m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu=0.5$ between block and surface of cube. $\left(g=10m/s^2\right)$



31. Force of friction acting on block is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q19

Circular Motion

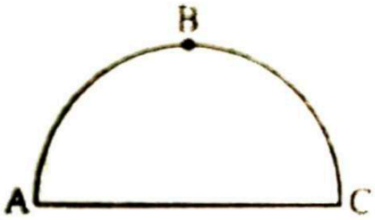
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **40%** got it right · Campus: **38%** (30 of 78)

Within reach

You: Wrong

19. If a particle starts from 'A' moving along the curved semi circular path as shown in figure with tangential acceleration 'a' then acceleration at B in magnitude is



A) $2a\left[\sqrt{1+\pi^2}\right]$ B) $a\left[\sqrt{1+\pi^2}\right]$ C) $a\left[\sqrt{\pi^2-1}\right]$ D) $a\pi\left[\sqrt{1+\pi^2}\right]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Work, Energy & Power – **Maximum spring extension – energy method**

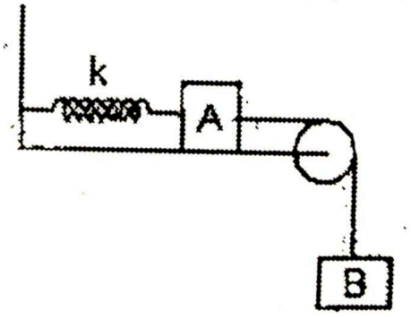
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **29%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Wrong

20. All the surfaces are smooth system is released from rest with spring unstretched. Then maximum extension in spring will be $(m_A = m; m_B = 2m)$



A) $\frac{mg}{K}$ B) $\frac{2mg}{K}$ C) $\frac{3mg}{K}$ D) $\frac{4mg}{K}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26

Circular Motion

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **24%** got it right · Campus: **17%** (13 of 78)

Tough one

You: Wrong

26. A particle of mass 2.5kg is moving in gravity free space with velocity $3\hat{i}m/s$. At $t = 0\text{sec}$ a force of magnitude 67.5N starts acting on the particle such that it is always perpendicular to instantaneous velocity. The minimum time after which particle has same velocity as the initial is $\frac{2\pi}{x}$ sec. The value of x .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Circular Motion – Net acceleration in a vertical circle at 37°

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **14%** (11 of 78)

Tough one

You: Wrong

29. A small ball of mass m is attached to the end of string of length $l = 1m$ whose other end is fixed. From its lowest position, the ball is given a kinetic energy $\frac{mgl}{5}$. Net acceleration $\left(in\ m/s^2\right)$ of ball at the instant when string makes an angle $\theta=37^o$ with vertical is ..... $\left(g = 10m/s^2\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Friction – Work by friction seen from an accelerating observer

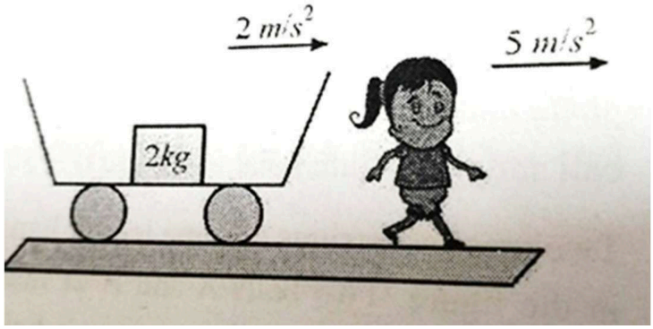
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **12%** (9 of 78)

Tough one

You: Wrong

30. An observer and a vehicle, both starts moving together from rest with acceleration $5m/s^2$ and $2m/s^2$ respectively. There is a 2 kg block on the floor of the vehicle and $\mu = 0.3$ between their surfaces. The work done by the frictional force on 2kg block as observed by running observer , during first 2seconds of motion is $\underline{-x}$ (J) $\left(g = 10m/s^2\right)$ find 'x'



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18

Friction – Static vs kinetic friction under a time-varying force

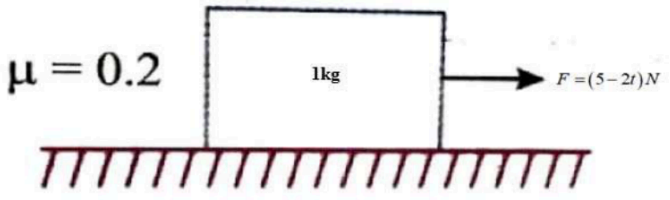
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **16%** got it right · Campus: **9%** (7 of 78)

Tough one

You: Wrong

18. The force acting on the block of mass 1 kg is given by $F = 5 - 2t$. The frictional force acting on the block after time $t = 2\text{sec}$ will be $\left(\mu = 0.2, g = 10 m/s^2\right)$



A) 2N B) 3N C) 1N D) zero

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

Work, Energy & Power – Work by conservative forces from the U-x graph

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **12%** got it right · Campus: **9%** (7 of 78)

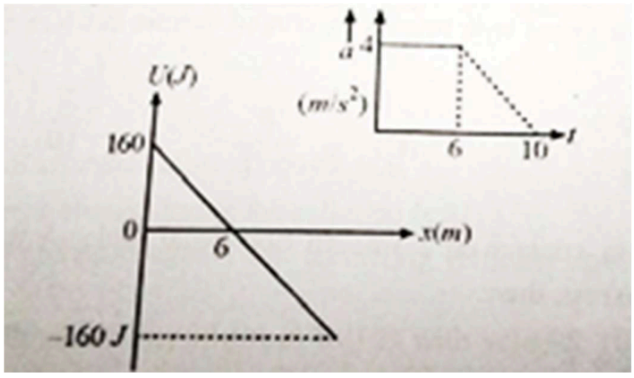
Tough one

You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



34. Work done by conservative forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Friction – Net work done by friction between sliding blocks

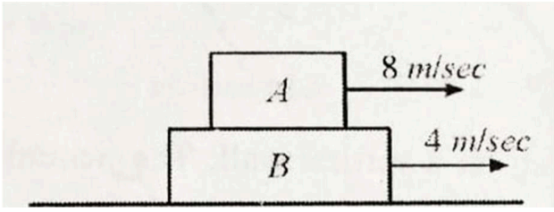
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **8%** got it right · Campus: **10%** (8 of 78)

Tough one

You: Skipped

27. Block A of mass 1kg is placed on rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given velocities as shown. If net workdone by frictional force is "-n" J. value of n is ……..



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Friction – Total force exerted by a block on an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

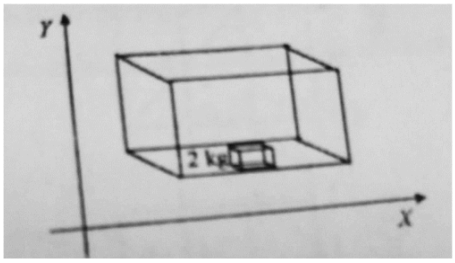
Top 100: **8%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Wrong

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V}=\left(5t\hat{i}+2t\hat{j}\right)m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu=0.5$ between block and surface of cube. $\left(g=10m/s^2\right)$



32. Total force exerted by block on cube is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

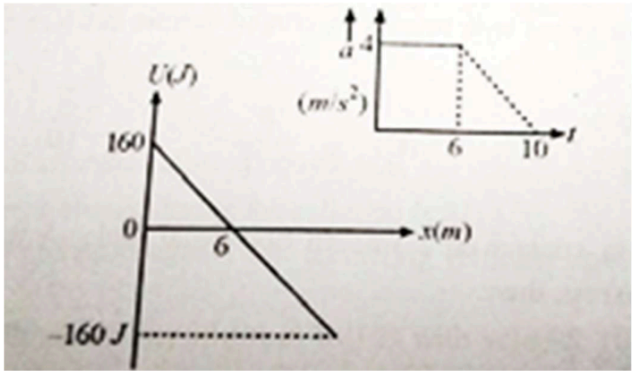
Q33 **Work, Energy & Power** – **Work-energy theorem from a-t and U-x graphs**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **2%** got it right · Campus: **1%** (1 of 78)

Tough one
You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



33. Work done by all forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Chemistry – questions to revisit

9 questions · easy misses first
cropped from the question paper

Q39 Hydrocarbons

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **87%** got it right · Campus: **69%** (54 of 78)

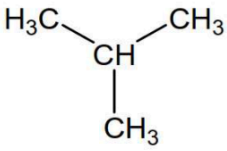
Easy miss
You: Wrong

39. Which of the following can be prepared by Wurtz reaction with best yield?

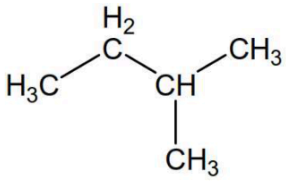
A) CH_3CH_3

B) $CH_3CH_2CH_2CH_3$

C)



D)



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

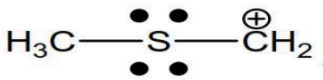
Q35 General Organic Chemistry

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4
Top 100: **58%** got it right · Campus: **42%** (33 of 78)

Within reach
You: Wrong

35. Which of the following cation would have greatest stability?

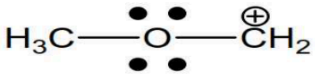
A)



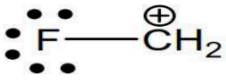
B)



C)



D)



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44 General Organic Chemistry

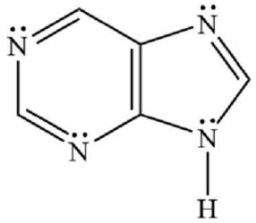
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **56%** got it right · Campus: **46%** (36 of 78)

Within reach

You: Wrong

44. The purine heterocycle occurs mainly in the structure of DNA. Identify number of 'N' atoms having localized lone pair of electrons.



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49 General Organic Chemistry

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **53%** got it right · Campus: **46%** (36 of 78)

Within reach

You: Wrong

PARAGRAPH-I

Electrophiles and nucleophiles are fundamental concepts of organic chemistry playing crucial roles in various reactions. Electron loving species that accept electron pairs to form covalent bonds are electrophiles, Nucleus loving species that donate electron pair to form covalent bonds.

49. How many of the following are nucleophiles

A) OR^- B) CH_3^+ C) NH_3 D) $AlCl_3$

E) NO_2^- F) NO_2^+ G) Br^+

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48

General Organic Chemistry – Counting electrophiles

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **37%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Wrong

PARAGRAPH-I

Electrophiles and nucleophiles are fundamental concepts of organic chemistry playing crucial roles in various reactions. Electron loving species that accept electron pairs to form covalent bonds are electrophiles, Nucleus loving species that donate electron pair to form covalent bonds.

48. How many of the following are electrophiles

A) Br^+ B) Na^+ C) CN^- D) BF_3

E) $BeCl_2$ F) NH_2^- G) SH^- H) OR^-

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q46

Isomerism – Count of optically active compounds

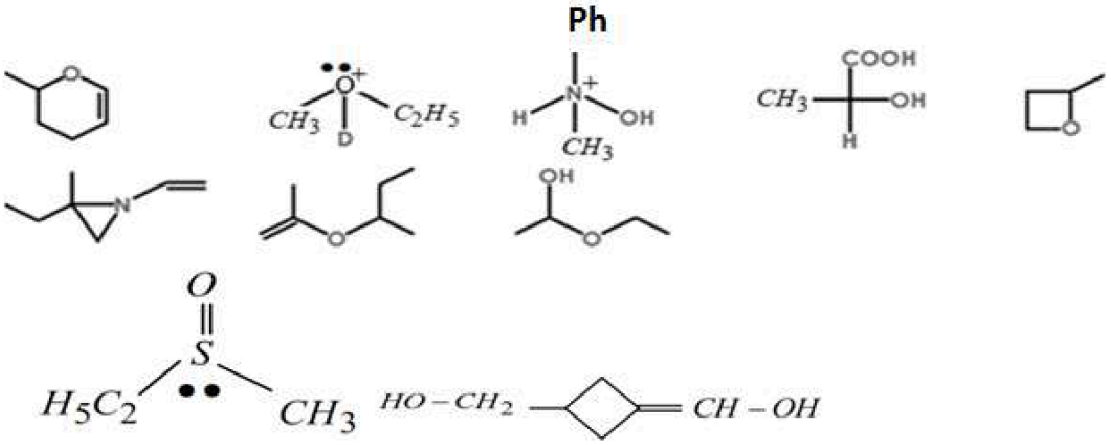
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **15%** got it right · Campus: **15%** (12 of 78)

Tough one

You: Wrong

46. How many given compounds are optically active



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51

Hydrocarbons – Chiral products of free-radical monobromination

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **4%** (3 of 78)

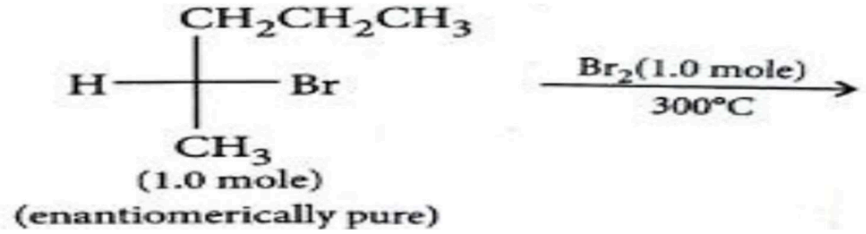
Tough one

You: Wrong

PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

51. In the following monobromination reaction, the number of possible chiral products is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Hydrocarbons – Dichlorination isomers of C_5H_{10} (VD 35)

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **7%** got it right · Campus: **5%** (4 of 78)

Tough one

You: Wrong

47. A hydrocarbon [X] is having vapour density 35 and it forms only 1 monochloro product under sunlight. How many isomeric products (including stereo) are possible when [X] undergoes dichlorination?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q50 **Hydrocarbons – Preparation of alkanes – count correct statements**
JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **4%** got it right · Campus: **3%** (2 of 78)

Tough one
You: Wrong

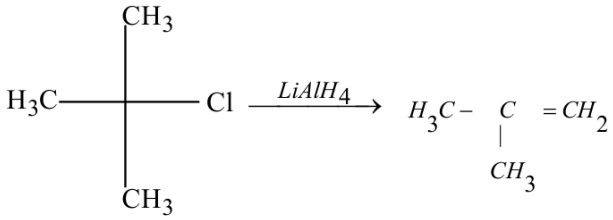
PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

50. Find the No of correct statements out of the following

A) Alkanes are prepared by catalytic hydrogenation of Alkenes where both hydrogen atoms add to the same face of the alkene (syn-addition).

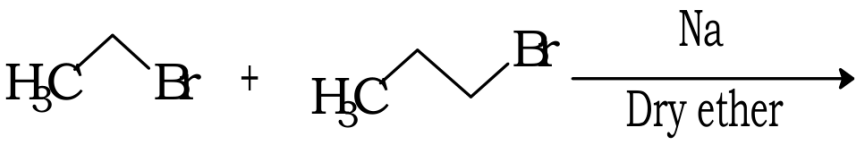
B) Alkyl halides when reacts with Lithium Aluminium Hydride form corresponding alkane: $R-X \xrightarrow{LiAlH_4} R-H$ for R : methyl , $1^0, 2^0$



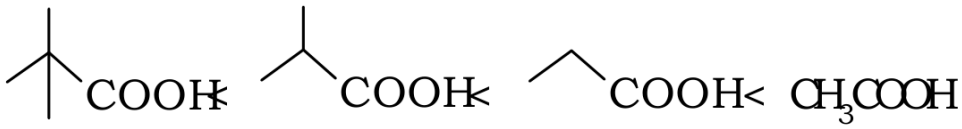
C)In case of Tertiary Alkyl halide is observed.

D) Grignard's reagent when treated with a polar protic solvent gives alkane:
$R-Mg+HA \rightarrow R-H+MgX(A)$

E) The different compounds obtained by the following reaction are 7 (only organic compounds)



F) Rate of sodalime decarboxylation for the following acids follow:



G) The side products of potassium butanoate during Kolbe's Electrolysis are propane, propene and propyl butanoate

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q58 — JEE Advanced: Q45. **Program rank** = rank in Pragya G9 (EXT) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 8 marks went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.